Advanced Data Modelling and Form Design

Aim

To extend the **Patient Appointment Booking Form** in Zoho Creator by implementing advanced field types and features such as lookup fields, image and file uploads, decimal and percentage fields, sub forms, URL fields, audio and video uploads, fee calculation, rich text content, digital signatures, auto-generated appointment IDs, and QR codes.

Theory

Advanced data modelling in Zoho Creator involves using specialized field types and advanced form components to collect, manage, validate, and automate detailed patient appointment information.

In this experiment, the basic **Patient Appointment Booking Form** is extended with advanced sections. These sections demonstrate the use of lookup fields, image and file uploads, decimal fields, sub forms, URL fields, audio and video uploads, decision boxes, currency and percentage fields, formula fields, rich text fields, digital signatures, date-time fields, auto number fields, and QR/barcode fields.

The advanced form contains the following sections:

- **Section E — Doctor Verification & Vitals**
- **Section F — Family Registration & Media Details**
- **Section G — Fee & Transport Preference**
- **Section H — Consent & System-Generated Values**

Section E — Doctor Verification & Vitals

- Doctor Availability Lookup (Lookup field, auto-fills doctor specialization and available time slots)
  - Patient Photo (Image field)
- Previous Medical Report Upload (File Upload field)
- Body Temperature (Decimal field)

Section F — Family Registration & Media Details

- Accompanying Family Members (SubForm field, repeating rows)
- Previous Report Portal URL (URL field)
- Symptom Video (Video field, optional)

- Symptom Voice Note (Audio field, optional)

Section G — Fee & Transport Preference

- Requires Ambulance Pickup (Decision Box)
- Consultation Fee (Currency field)
- Insurance Coverage Percent (Percent field)
- Net Payable Fee (Formula field, read-only, auto-computed)

Section H — Consent & System-Generated Values

- Clinic Guidelines Display (Rich Text field)
- Patient Digital Signature (Signature field)
- Appointment Submitted On (Date-Time field)
- Appointment ID (Auto Number field)
- Appointment QR Code (QR/Barcode field)

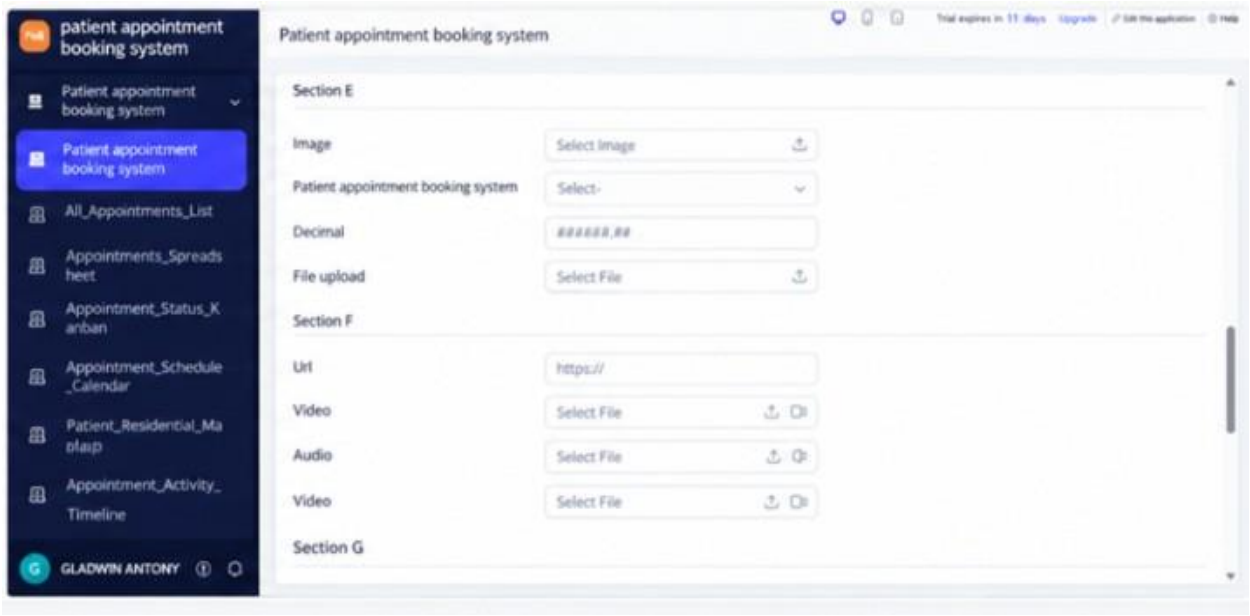
Procedure

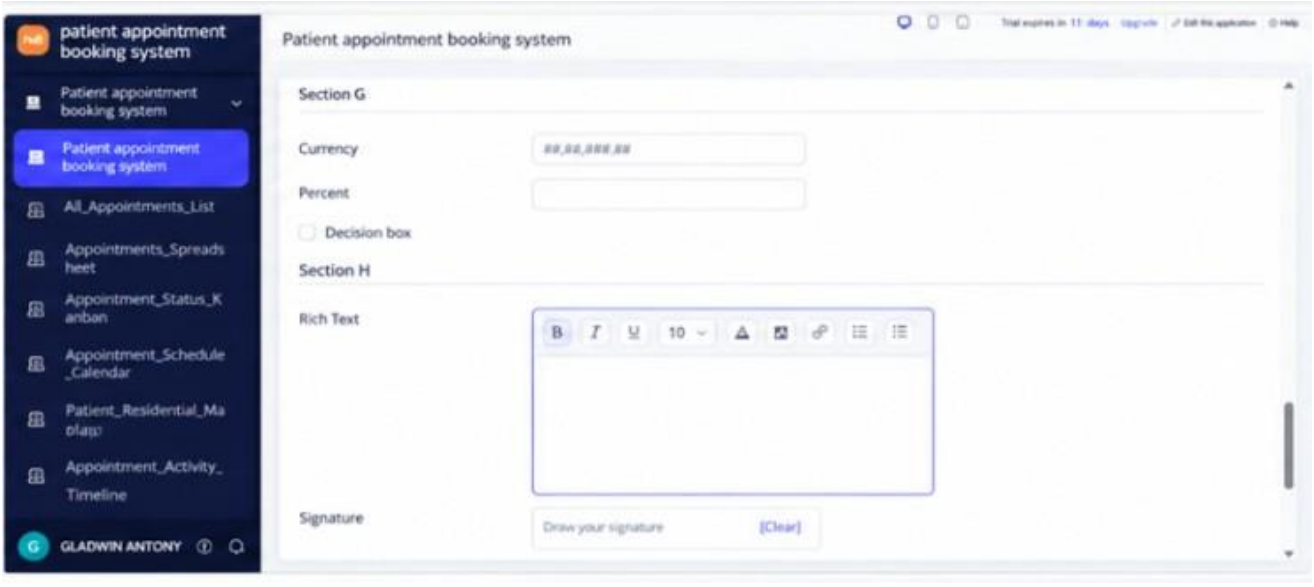
1. Open the existing **Patient Appointment Booking Form** created in Zoho Creator.
2. Add **Section E – Doctor Verification & Vitals**.
3. Add the **Doctor Availability Lookup** field.
4. Add the **Patient Photo** image upload field.
5. Add the **Previous Medical Report Upload** file field.
6. Add the **Body Temperature** decimal field.
7. Create **Section F – Family Registration & Media Details**.
8. Add a **SubForm** for accompanying family member details.
9. Configure Family Member Name, Relation, and Age inside the SubForm.
10. Add the **Previous Report Portal URL** field.
11. Add the **Symptom Video** upload field.
12. Add the **Symptom Voice Note** audio field.
13. Create **Section G – Fee & Transport Preference**.
14. Add the **Requires Ambulance Pickup** decision box.
15. Add the **Consultation Fee** currency field.
16. Add the **Insurance Coverage Percent** field.
17. Add the **Net Payable Fee** formula field.
18. Create **Section H – Consent & System-Generated Values**.

19. Add the **Clinic Guidelines Display** rich text field.
20. Add the **Patient Digital Signature** field.
21. Add the **Appointment Submitted On** date-time field.
22. Add the **Appointment ID** auto-number field.
23. Add the **Appointment QR Code** field.
24. Configure the required advanced field properties and validations.
25. Save and preview the completed form.
26. Enter sample patient data into the advanced fields.
27. Add sample family members through the SubForm.
28. Upload the required patient photo and previous medical report.
29. Provide the URL, media, ambulance, fee, insurance, declaration, and signature information.
30. Submit the completed form.
31. Verify the submitted information in the **Appointment_Booking_Report**.

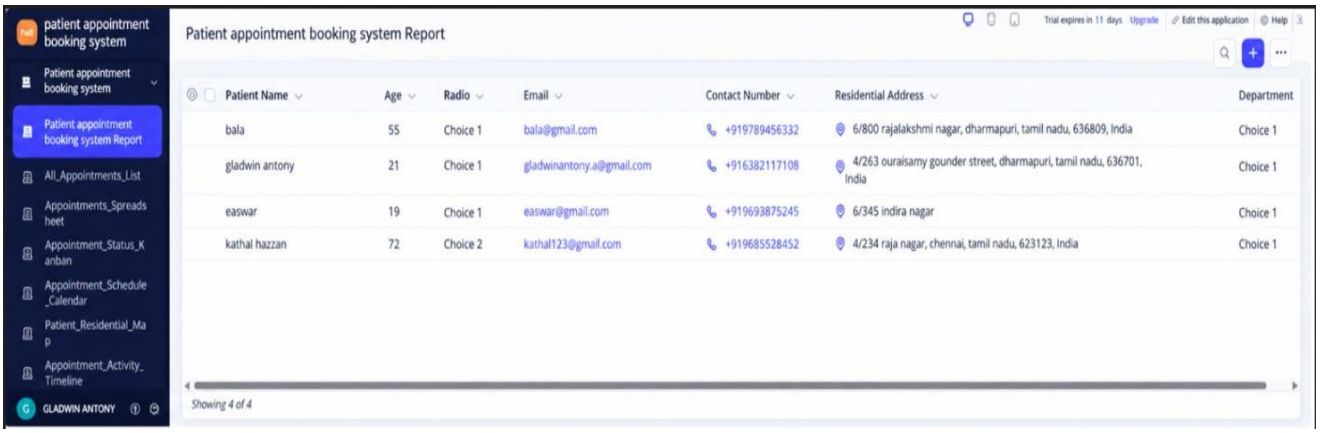
Screenshots:

Patient Appointment Booking Form:





Patient Appointment Booking Report:



Result

The **Patient Appointment Booking Form** was successfully extended in Zoho Creator with advanced fields, appropriate validations, automated fee calculation, doctor availability lookup, patient document and media uploads, digital signature, and system-generated appointment ID and QR code. The submitted patient records were successfully stored and displayed in the **Appointment_Booking_Report**, enabling efficient appointment verification, payment checking, and reception check-in.